PROJECT REPORT

EXPERIMENT - 1

Project title

IoT-Based Smart Irrigation and Water Management System

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IoT-Based Smart Irrigation and Water Management System

Project Overview

The IoT-Based Smart Irrigation and Water Management System Using ESP8266 is an automated irrigation system designed to provide water to plants based on their actual soil moisture conditions. The system uses a soil moisture sensor to continuously monitor the moisture level of the soil and a DHT11 sensor to measure temperature and humidity.

The ESP8266 Node MCU acts as the main controller. When the soil becomes dry below a predefined moisture level, the ESP8266 activates the relay module, which switches ON the mini water pump. The pump supplies water to the plant until the soil reaches the required moisture level, after which the ESP8266 switches the pump OFF.

This automatic process helps reduce water wastage, avoid over-irrigation, minimize manual effort, and improve efficient water management in agricultural applications.

Main Objectives:

- To automate the irrigation process based on soil moisture conditions.
- To monitor soil moisture continuously using a soil moisture sensor.
- To measure temperature and humidity using the DHT11 sensor.
- To control the water pump automatically using the ESP8266 and relay module.
- To reduce unnecessary water consumption during irrigation.
- To prevent over-irrigation and under-irrigation of plants.
- To minimize manual effort required for regular watering.
- To improve efficient water management in agricultural applications.
- To develop a simple and low-cost smart irrigation system using readily available components.
- To provide a reliable automatic irrigation solution for small-scale agricultural and gardening

Problem Statement

Traditional irrigation methods often depend on manual monitoring and fixed watering schedules, which can lead to over-irrigation, under-irrigation, and unnecessary water consumption. Farmers or users may not always be able to continuously monitor the moisture condition of the soil and surrounding environmental conditions.

To overcome these challenges, an IoT-Based Smart Irrigation and Water Management System Using ESP8266 is proposed. The system continuously monitors soil moisture, temperature, and humidity using suitable sensors. Based on the soil moisture condition, the ESP8266 automatically controls a water pump through a relay module. When the soil becomes dry, the pump is activated to provide water, and when sufficient moisture is reached, the pump is switched OFF.

The main aim is to automate irrigation, reduce water wastage, minimize manual effort, and provide efficient water management for agricultural applications.

Problems Addressed:

The proposed IoT-Based Smart Irrigation and Water Management System Using ESP8266 addresses the following problems:

- **Manual Irrigation** – Reduces the need for users to manually switch the water pump ON and OFF.
- **Water Wastage** – Prevents unnecessary watering by supplying water only when the soil requires it.
- **Over-Irrigation** – Helps avoid excessive watering that can negatively affect plant growth.
- **Under-Irrigation** – Ensures plants receive water when the soil becomes sufficiently dry.
- **Lack of Soil Monitoring** – Continuously monitors soil moisture using the soil moisture sensor.
- **Fixed Irrigation Schedules** – Replaces fixed watering schedules with moisture-based automatic irrigation.
- **Environmental Monitoring** – Measures temperature and humidity using the DHT11 sensor to understand surrounding conditions.

- **Human Effort** – Minimizes the need for continuous supervision of the irrigation process.
- **Inefficient Water Management** – Improves the efficient use of available water in agricultural and gardening applications.
- **Lack of Automation** – Provides an automatic control mechanism using the ESP8266, relay module, and water pump.

Proposed Solution

The proposed solution is to develop an IoT-Based Smart Irrigation and Water Management System Using ESP8266 that automatically controls irrigation based on the actual condition of the soil.

- Develop an automatic irrigation system using ESP8266 Node MCU.
- Monitor soil moisture using the soil moisture sensor.
- Measure temperature and humidity using the DHT11 sensor.
- Use the ESP8266 to process sensor readings.
- Turn the water pump ON through the relay when the soil is dry.
- Turn the pump OFF automatically when sufficient moisture is reached.
- Reduce water wastage and manual effort.
- Improve efficient water management in agricultural applications.

Proposed Operation:

- The soil moisture sensor continuously detects the moisture level of the soil.
- The DHT11 sensor measures temperature and humidity.
- The ESP8266 Node MCU receives and processes the sensor readings.
- If the soil is dry, the ESP8266 activates the relay.
- The relay turns ON the water pump.
- Water is supplied to the plants until the required moisture level is reached.
- When the soil becomes sufficiently moist, the ESP8266 deactivates the relay.
- The water pump turns OFF automatically.

- This cycle repeats continuously to maintain suitable soil moisture.
- The system helps save water and reduce manual irrigation effort.

System Architecture

Input Section:

- Soil Moisture Sensor – Detects the moisture level of the soil.
- DHT11 Sensor – Measures temperature and humidity.

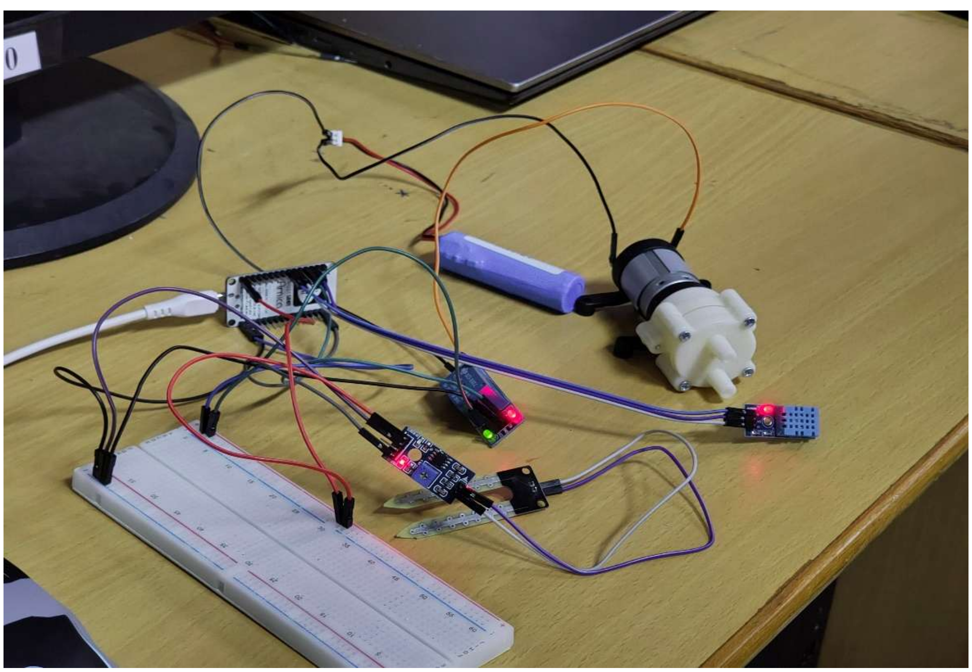
Processing & Control Section:

- ESP8266 Node MCU – Receives sensor data, processes it, and decides whether irrigation is required.
- Relay Module – Acts as an electronic switch to control the water pump.

Output Section:

- Mini Water Pump – Supplies water to the plants when the soil is dry.

Image:



Circuit table

Component	Pin	ESP8266 Node MCU Connection
Soil Moisture Sensor	VCC	3.3V
	GND	GND
	A0	A0
DHT11 Sensor	VCC	3.3V
	GND	GND
	DATA	D4 (GPIO2)
Relay Module	VCC	5V / VIN
	GND	GND
	IN	D1 (GPIO5)
Water Pump	Positive	Relay NO/COM through suitable external supply
	Negative	External supply GND

Important Pin Mapping:

Soil Moisture Sensor:

- AO → A0
- VCC → 3.3V
- GND → GND

DHT11 Sensor:

- DATA → D4 (GPIO2)
- VCC → 3.3V
- GND → GND

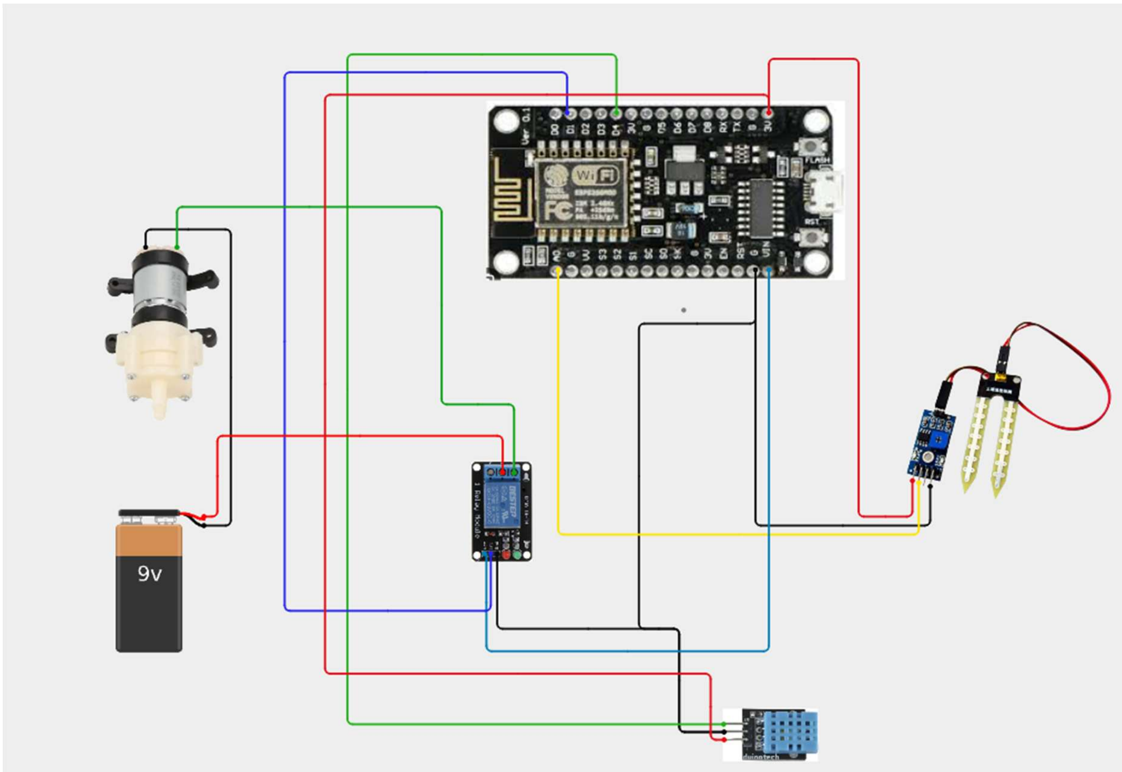
Relay Module:

- IN → D1 (GPIO5)
- VCC → Supply
- GND → GND

Mini Water Pump:

- Connected through the relay.
- Use a suitable external power supply for the pump.

Circuit Design



Algorithm

Step 1: Start the system.

Step 2: Initialize the ESP8266, Soil Moisture Sensor, DHT11, and Relay Module.

Step 3: Read the soil moisture value.

Step 4: Read temperature and humidity from the DHT11 sensor.

Step 5: Compare the soil moisture value with the preset threshold.

Step 6: If the soil is dry, send a signal from ESP8266 to the relay.

Step 7: Relay turns ON the water pump.

Step 8: Water is supplied to the plant.

Step 9: Continuously monitor the soil moisture.

Step 10: When the soil reaches the required moisture level, turn the relay OFF.

Step 11: Water pump stops automatically.

Step 12: Repeat the process continuously

Coding

```
#include <ESP8266WiFi.h>
#include <ThingSpeak.h>
#include <DHT.h>

//================ WiFi Credentials ====================
const char* ssid = "Rithick";
const char* password = "12345678";

//================ ThingSpeak ===========================
unsigned long channelID = 3455765;
const char* writeAPIKey = "9JHQB4M5U1LNNLJQ";

WiFiClient client;

//================ Pin Definitions ====================
#define DHTPIN 16           // GPIO2
#define DHTTYPE DHT11

#define SOIL_PIN A0
#define RELAY_PIN 00        // GPIO5
```

```
DHT dht(DHTPIN, DHTTYPE);

//================ Variables ===========================
int soilValue = 0;
float temperature = 0;
float humidity = 0;

int dryThreshold = 600;

bool pumpStatus = false;

//====================================================

void connectWiFi()
{
  Serial.print("Connecting to WiFi");

  WiFi.mode(WIFI_STA);
  WiFi.begin(ssid, password);

  int attempts = 0;

  while (WiFi.status() != WL_CONNECTED && attempts < 30)
  {
    delay(500);
    Serial.print(".");
```

```
    attempts++;
  }

  if (WiFi.status() == WL_CONNECTED)
  {
    Serial.println();
    Serial.println("WiFi Connected");
    Serial.print("IP Address: ");
    Serial.println(WiFi.localIP());
  }
  else
  {
    Serial.println();
    Serial.println("WiFi Connection Failed!");
  }
}

//========================================================

void setup()
{
  Serial.begin(9600);

  pinMode(RELAY_PIN, OUTPUT);

  // Relay OFF (Active LOW Relay)
```

```
  digitalWrite(RELAY_PIN, HIGH);

  dht.begin();

  connectWiFi();

  ThingSpeak.begin(client);

  Serial.println();
  Serial.println("========================================");
  Serial.println(" SMART IRRIGATION SYSTEM");
  Serial.println("========================================");
}

//==========================================================

void loop()
{
  // Reconnect WiFi if disconnected
  if (WiFi.status() != WL_CONNECTED)
  {
    Serial.println("WiFi Lost! Reconnecting...");
    connectWiFi();
  }

  // Read Soil Moisture
```

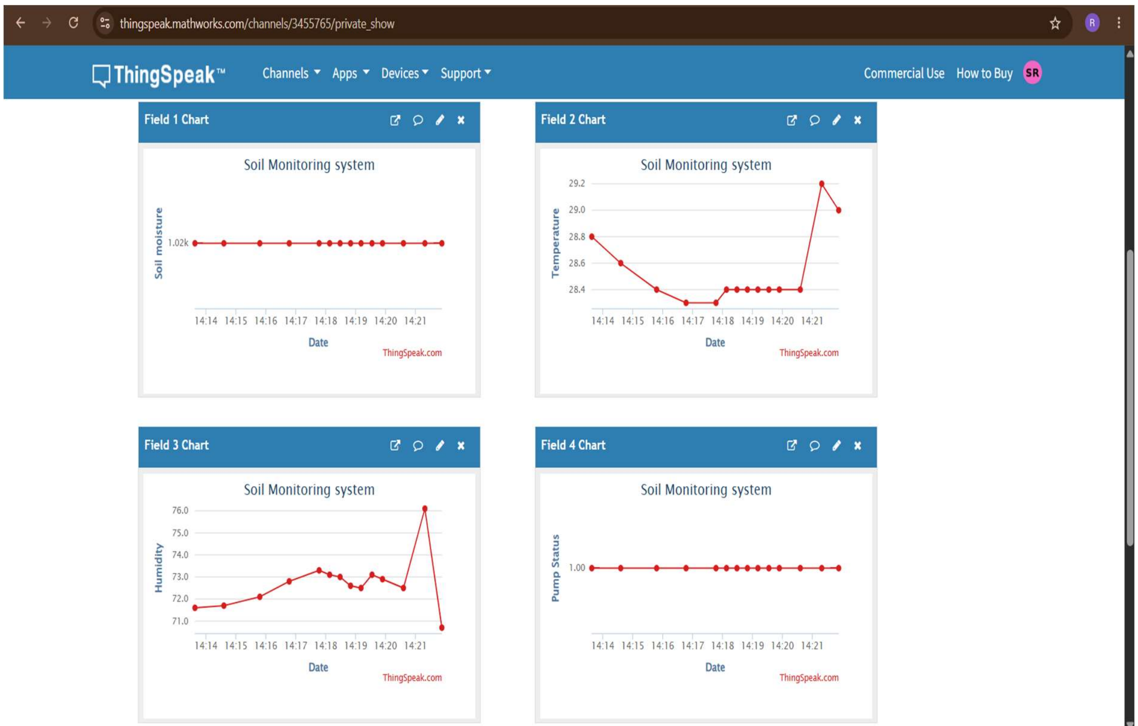
```
soilValue = analogRead(SOIL_PIN);

// Read DHT11
temperature = dht.readTemperature();
humidity = dht.readHumidity();

if (isnan(temperature) || isnan(humidity))
{
  Serial.println("DHT11 Read Failed!");
  delay(2000);
  return;
}

// Pump Control
if (soilValue > dryThreshold)
{
  digitalWrite(RELAY_PIN, LOW);     // Pump ON
  pumpStatus = true;
}
else
{
  digitalWrite(RELAY_PIN, HIGH);    // Pump OFF
  pumpStatus = false;
}

// Display Data
```

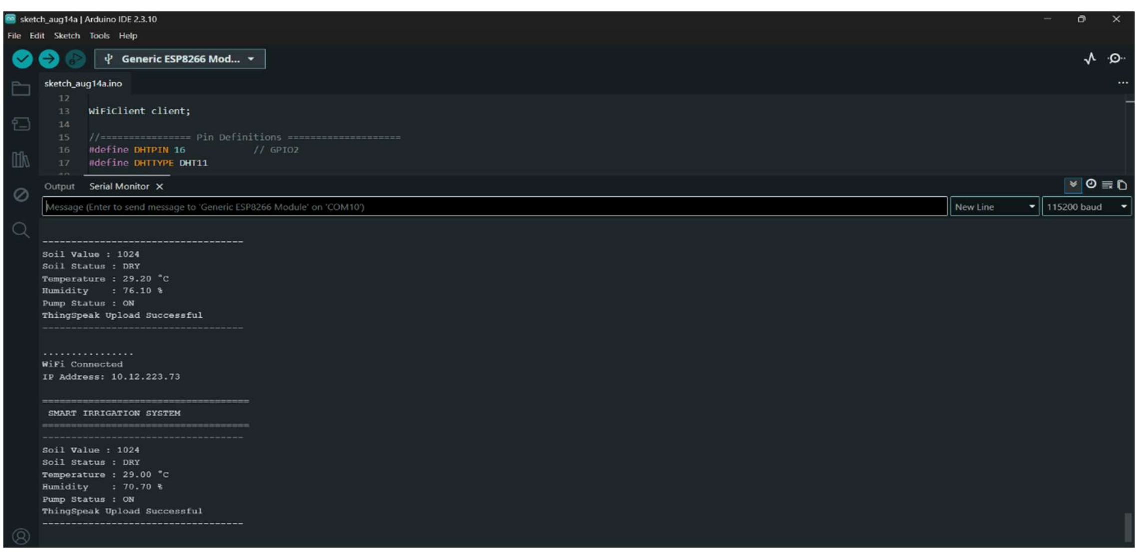
```
Serial.println("------------------------------------");

Serial.print("Temperature : ");
Serial.print(temperature);
Serial.println(" °C");

Serial.print("Humidity    : ");
Serial.print(humidity);
Serial.println(" %");

Serial.print("Soil Value  : ");
Serial.println(soilValue);

if (soilValue > dryThreshold)
  Serial.println("Soil Status : DRY");
else
  Serial.println("Soil Status : WET");

Serial.print("Pump Status : ");

if (pumpStatus)
  Serial.println("ON");
else
  Serial.println("OFF");

// Upload to ThingSpeak
```

```
if (WiFi.status() == WL_CONNECTED)
{
 ThingSpeak.setField(1, soilValue);
 ThingSpeak.setField(2, temperature);
 ThingSpeak.setField(3, humidity);
 ThingSpeak.setField(4, pumpStatus);

 int response = ThingSpeak.writeFields(channelID, writeAPIKey);

 if (response == 200)
 {
  Serial.println("ThingSpeak Upload Successful");
 }
 else
 {
  Serial.print("ThingSpeak Error Code: ");
  Serial.println(response);
 }
}
else
{
 Serial.println("No WiFi - Data Not Uploaded");
}

Serial.println("------------------------------------");
Serial.println();
```

```
  delay(20000);   // ThingSpeak minimum update interval
}
```

Coding output:





System Working

Step 1: The system is powered ON and the ESP8266 initializes all connected sensors and the relay.

Step 2: The Soil Moisture Sensor detects the moisture level of the soil.

Step 3: The DHT11 Sensor measures the surrounding temperature and humidity.

Step 4: The ESP8266 receives and processes the sensor readings.

Step 5: The ESP8266 compares the soil moisture reading with the predefined threshold.

Step 6: If the soil is dry, the ESP8266 sends a signal to the relay.

Step 7: The relay switches ON the water pump.

Step 8: The pump supplies water to the plants.

Step 9: The soil moisture sensor continuously checks the moisture level.

Step 10: When the soil reaches the required moisture level, the ESP8266 switches the relay OFF.

Step 11: The water pump stops automatically.

Step 12: The system continuously repeats this process to maintain proper soil moisture and reduce water wastage.

Bill of Materials

S. No.	Component	Quantity
1	ESP8266 NodeMCU	1
2	Soil Moisture Sensor	1
3	DHT11 Temperature & Humidity Sensor	1
4	Relay Module	1
5	Mini Water Pump	1
6	Breadboard	1
7	Jumper Wires	As required
8	Suitable External Power Supply for Pump	1

PROTOTYPE IMPLEMENTATION

The Smart Irrigation and Water Management System is implemented using ESP8266 Node MCU, Soil Moisture Sensor, DHT11, Relay Module, and Mini Water Pump.

- Hardware Assembly: Connect the soil moisture sensor and DHT11 to the ESP8266 for sensing soil and environmental conditions.
- Pump Control: Connect the mini water pump to the relay module. The ESP8266 controls the relay according to soil moisture.
- Sensor Monitoring: The soil moisture sensor continuously measures the moisture level, while DHT11 measures temperature and humidity.
- Automatic Irrigation: When the soil becomes dry, the ESP8266 activates the relay and turns ON the water pump.
- Water Supply: The pump supplies water to the plant until the required moisture level is reached.
- Pump OFF: When adequate moisture is detected, the ESP8266 turns OFF the relay and pump.
- Testing: The prototype is tested with dry and wet soil to verify automatic pump operation.

Hardware Setup:

The prototype hardware is assembled as follows:

- **ESP8266 Node MCU** – Main controller for reading sensors and controlling the pump.
- **Soil Moisture Sensor** – Inserted into the soil to measure moisture level.
- **DHT11 Sensor** – Measures temperature and humidity.
- **Relay Module** – Acts as an electronic switch to control the water pump.
- **Mini Water Pump** – Pumps water to the plant when the soil is dry.
- **Breadboard & Jumper Wires** – Used to make the circuit connections.
- **Power Supply** – Provides power to the Node MCU and pump circuit.

Prototype Working:

- The soil moisture sensor detects the moisture level of the soil.
- The ESP8266 Node MCU reads the sensor value and checks whether the soil is dry or wet.
- If the soil is dry, the ESP8266 activates the relay module.
- The relay turns ON the mini water pump and water is supplied to the plant.
- When the soil reaches the required moisture level, the ESP8266 deactivates the relay.
- The water pump turns OFF, preventing unnecessary water usage.
- The DHT11 sensor continuously monitors temperature and humidity.
- This process repeats automatically to maintain proper soil moisture.

Expected Prototype Output:

Condition	Soil Moisture	DHT11 Reading	Pump Status	System Action
Dry Soil	Low	Temperature & Humidity detected	ON	Irrigation starts
Moderate Moisture	Medium	Temperature & Humidity detected	ON/OFF	Moisture monitored
Wet Soil	High	Temperature & Humidity detected	OFF	Irrigation stops
Very Wet Soil	Very High	Temperature & Humidity detected	OFF	Water is conserved
Normal Environment	Normal	Temperature & Humidity detected	Based on soil moisture	Continuous monitoring

Prototype Demonstration:

The prototype is powered ON, and the ESP8266 initializes the soil moisture sensor and DHT11 sensor. The soil moisture sensor is placed in dry soil, and the ESP8266 detects the low moisture level. It then activates the relay, which turns ON the mini water pump and supplies water to the plant. As the soil becomes sufficiently wet, the moisture value increases, causing the ESP8266 to deactivate the relay and turn OFF the pump. Meanwhile, the DHT11 continuously measures temperature and humidity. This demonstration verifies the automatic irrigation and water-saving operation of the prototype.

Results & Performance Analysis

The developed Smart Irrigation System successfully detects soil moisture and automatically controls the water pump. When the soil becomes dry, the ESP8266 activates the relay and starts irrigation. When sufficient moisture is reached, the pump is automatically switched OFF. The DHT11 also provides temperature and humidity monitoring. The prototype reduces unnecessary watering, provides reliable automatic operation, and helps conserve water while maintaining suitable soil moisture for plants.

Expected Results:

- Automatically detects the soil moisture level.
- Turns ON the water pump when the soil is dry.
- Turns OFF the water pump when sufficient moisture is reached.
- Continuously monitors temperature and humidity using DHT11.
- Reduces unnecessary water consumption.
- Minimizes manual irrigation effort.
- Maintains suitable moisture conditions for plants.
- Provides reliable and efficient automatic irrigation.

Performance Analysis

Parameter	Performance	Observation
Soil Moisture Detection	Good	Detects dry and wet soil conditions effectively
Automatic Irrigation	Good	Pump operates automatically based on soil moisture
Pump Control	Reliable	Relay switches the pump ON/OFF as required
Temperature Monitoring	Good	DHT11 measures temperature continuously
Humidity Monitoring	Good	DHT11 provides humidity readings
Water Conservation	Effective	Reduces unnecessary water usage
Response Time	Fast	Pump responds quickly to moisture changes
Overall System Performance	Satisfactory	Provides reliable automatic irrigation